

Alexandre Kaizeler

Computational Biologist | Lisbon, Portugal

alexandre.kaizeler.afonso@gmail.com | [linkedin.com/in/alexandre-kaizeler/](https://www.linkedin.com/in/alexandre-kaizeler/) | github.com/A-Kaizeler-A | [Website](#)

SUMMARY

I am an ambitious computational biologist with 4+ years of experience analyzing large-scale, high-dimensional biological datasets and translating complex findings into clear, decision-ready scientific narratives. As first and corresponding author, I led a reanalysis of public transcriptomic data, rigorously separating genuine biological signal from technical artifacts. My computational work has directly contributed to identifying a druggable dependency in cancer, and I built public tools ([voyAGER](#), [CHARM](#)) with the goal of making advanced statistical analyses interpretable to non-expert audiences. That same ambition pushes me to hold my own work to a high bar, revisiting and refining it until it meets that standard. I bring peer-reviewed publication experience, rigorous quantitative training, and strong written communication skills to cross-disciplinary scientific work.

LANGUAGES

Portuguese — Native | English — Professional Working Proficiency (C2)

TECHNICAL SKILLS

Programming	R (expert), Python (advanced), JavaScript (familiar)
Machine Learning and Statistics	Statistical modeling, hypothesis generation and testing, dimensionality reduction (PCA), classification (Elastic Net, Sparse Centroids)
Genomics and Sequencing Analysis	Differential expression analysis, differential splicing analysis, gene set enrichment analysis (GSEA), variant calling, mutational analysis, NGS quality control (FastQC, MultiQC), read depth and library complexity assessment, DSL (Nextflow)
Data Visualization	R Shiny, plotly, ggplot2, interactive tables and plots, dashboard/UI design for non-technical end users
Scientific Communication and Review	Peer-reviewed manuscript authorship (corresponding and co-author), formal software peer review (Bioconductor), scientific writing, data quality evaluation and curation, cross-disciplinary and asynchronous collaboration

SOFT SKILLS

Critical Thinking	Identified a hidden methodological bias in public stress granule transcriptomic data that had gone unnoticed in prior studies, leading to a novel, published finding. I enjoy this kind of problem-solving process across every project I take on.
Collaboration and Teamwork	Co-authored several peer-reviewed publications with multi-lab, multi-institution teams, including as first and corresponding author.
Scientific Communication	Built public tools (voyAGER, CHARM), authored peer-reviewed manuscripts, and regularly presented posters and talks at GIMM's institutional research days, translating complex analyses for both specialist and non-specialist audiences.
Adaptability and Self-Directed Learning	Transitioned from wet-lab cell biology (MSc) to a full computational biology stack (R, Python, statistical modeling) mainly through self-directed learning.

PROFESSIONAL EXPERIENCE

Computational Biologist / PhD Researcher

Gulbenkian Institute for Molecular Medicine (GIMM) / Instituto de Medicina Molecular, University of Lisbon — Lisbon, Portugal | February 2022 – Present

- Led a large-scale reanalysis of 16 independent public human stress granule transcriptomic datasets, assessing sequencing quality (FastQC, MultiQC) and read depth and library complexity across samples, and correcting for a previously unrecognized length-based technical bias in one isolation method. Mitochondrial RNA enrichment in SGs was identified as a reproducible signal with a proposed role in immune signaling, and a reusable

framework was established for separating genuine biology from methodological artifacts in the field. The experience reinforced a habit I now apply by default: treating an interesting pattern as a hypothesis to test, and not take as proven. [GitHub Repo](#).

- Co-led development of voyAGER, a public web application built on a novel bioinformatics pipeline processing RNA-sequencing data from 900+ individuals across 49 tissues (from GTEx). The statistical backend and interactive front-end (visualization, filtering, and export tools) were built to let non-technical users query and interpret results independently. This project offered me a sharper sense of how to make advanced statistics usable by people without a statistics background. [GitHub Repo](#).
- Designed and built an interactive R Shiny analytics platform (CHARM) for similarity search, correlation analysis, and GSEA-based pathway enrichment of RNA-binding proteins, engineering the data layer for performance at scale through precomputed similarity matrices, lazy-loading, and serialized caching. Guardrails such as runtime warnings were added to guide users toward faster queries. Working through these performance constraints gave me a much better intuition for scalable engineering data systems. [GitHub Repo](#).
- Contributed to markeR, an open-source R package providing a modular framework for systematic gene set-level analysis of transcriptomic data. I built and tested package functionality and contributed to the documentation and code that passed Bioconductor's formal review process for public release. The package has since been published on Bioconductor, giving the broader research community a reusable tool rather than one-off analysis scripts. Passing a formal peer review process instilled a lasting discipline around writing code and documentation built for other people to trust. [Bioconductor Page](#).

Research Intern, MSc Thesis

Champalimaud Foundation — Lisbon, Portugal | 2020 – 2021

- I investigated cell competition using a colorectal cancer co-culture model. I ran RNA-Seq differential expression analysis comparing winner and loser cells. Candidate genes were validated by RT-qPCR and tested functionally through gene knockdowns, identifying three genes required for loser-cell elimination. I also linked cell competition to cell migration and chemotherapy response for the first time in this cancer type. It was my first time carrying a project end-to-end, from a hypothesis to *in silico* analysis and its wet-lab validation, and it shaped how I still approach projects today.

EDUCATION

PhD, Biomedical Sciences — University of Lisbon (Feb 2022 – Present)

MSc, Oncobiology — University of Lisbon (2018 – 2021)

BSc, Biomedical Sciences — University of Algarve (2015 – 2018)

SELECTED PUBLICATIONS/IMPACT

- **voyAGER:** Public web tool for exploring age-related gene expression across 49 human tissues; co-developed statistical engine and interactive UI. Schneider, A. L. *, Martins-Silva, R. *, **Kaizeler, A. ***, Saraiva-Agostinho, N., & Barbosa-Morais, N. L. (2024). *voyAGER, a free web interface for the analysis of age-related gene expression alterations in human tissues*. *Elife*, 12, RP88623. * First Author.
 - **markeR:** Open-source R package for systematic gene set-level analysis of transcriptomic data; contributed code and testing that passed formal peer review for public software release. Martins-Silva, R., **Kaizeler, A.**, & Barbosa-Morais, N. L. (2026). *Exploring molecular signatures of senescence with marker, an R Toolkit for evaluating gene sets as phenotypic markers*. *NAR Genomics and Bioinformatics*, 8(2), lqag057.
 - **CHARM:** R Shiny tool for large-scale similarity search and pathway enrichment analysis, engineered for performance on multi-gigabyte datasets. Directly enabled the identification of a druggable spliceosome-related dependency in RAS-driven cancer. Wagner, V., Bousset, L., Ascensão-Ferreira, M., Sun, B., Barragan Avila, J. E., **Kaizeler, A.**, et al. (2026). *Spliceosome induction is a druggable dependency of RAS-driven senescence and cancer*. *Nature Communications*.
 - **Immune and tumoral-related analyses in T-ALL:** Performed differential expression analysis, GSEA and variant calling/mutational analysis in a murine model of spontaneous T-ALL. Oliveira, B. S., Paiva, R. A., Pinto, C. C., Paiva, R. V., Pereira, M. X., Ramos, C. V., et al. (2025). *Limited bone marrow chimerism impairs cell competition in the thymus and causes leukemia*. *bioRxiv*, 2025-12.
 - **Review of two tumor-related proteins, TP53 and SPARC** – Performed a bibliographical review of the impact of these two proteins in tumors. Camacho, D., Jesus, J. P., Palma, A. M., Martins, S. A., **Kaizeler, A.**, Peixoto, M. L., et al. (2020). *SPARC-p53: The double agents of cancer*. *Advances in cancer research*, 148, 171-199.
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